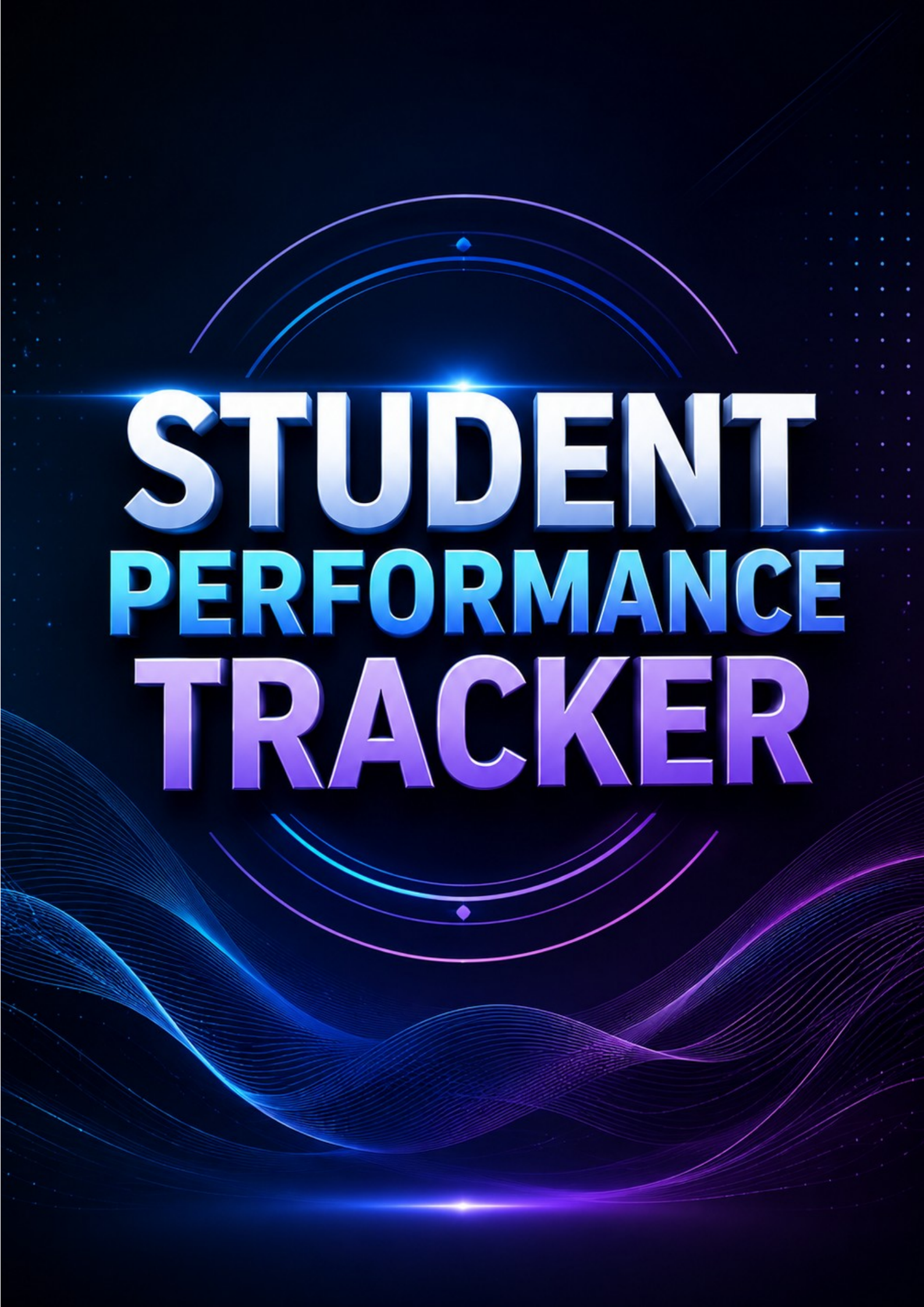




STUDENT PERFORMANCE TRACKER

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1. INTRODUCTION

Project Overview

Student Performance Tracker is a Flask-based web application designed to manage student academic records. The application allows users to add students, record subject-wise marks, view student information, calculate averages, identify top-performing students, and manage records through editing and deletion.

2. TECHNICAL ARCHITECTURE

Technology Stack

Frontend:

- HTML
- CSS

Backend:

- Python
- Flask

Database:

- SQLite

Deployment:

- Render

Version Control:

- GitHub
-

3. APPLICATION INTERFACE

Features of the Application

- Add Student

- Add Grades
- View Student Records
- Search Students
- Edit Grades
- Delete Grades
- Student Performance Reports
- Class Average Calculation
- Topper Identification
- Dashboard Statistics

Home Page

The Home Page serves as the dashboard of the application.

It provides quick access to:

1. Add Student
2. Add Grades
3. View Students
4. Class Average
5. Find Topper

The dashboard also displays statistics such as:

- Total Students
- Total Subjects
- Class Average
- Top Performer

4. CORE FUNCTIONALITIES

Add Student

Purpose: Register a new student in the system.

Steps:

1. Click "Add Student".
2. Enter Student Name.
3. Enter Roll Number.

4. Click "Add Student".

The system validates that the Roll Number is unique.

Add Grades

Purpose: Add subject-wise marks for a student.

Steps:

1. Click "Add Grades".
2. Enter Roll Number.
3. Enter Subject Name.
4. Enter Marks (0–100).
5. Click "Add Grade".

The application checks that the student exists before saving the marks.

View Students

Purpose: Display all student records.

Information shown:

- Roll Number
- Student Name
- Subject
- Marks

Additional actions available:

- View Report
- Edit Marks
- Delete Subject Record

Search functionality is available at the top of the page.

Class Average

Purpose: Display average marks for students.

The page automatically calculates average marks based on all recorded grades.

5. REPORTING & ANALYTICS

Student Report

Purpose: Display detailed information for a single student.

The report includes:

- Student Name
- Roll Number
- Subject-wise Marks
- Average Marks

To open a report:

1. Navigate to View Students.
2. Click "Report".

Find Topper

Purpose: Identify the highest-performing student(s).

The application calculates average marks for all students and displays the student(s) with the highest average.

If multiple students have the same highest average, all of them are displayed.

Edit Grade

Purpose: Modify an existing subject record.

Steps:

1. Open View Students.
2. Click "Edit".
3. Update Subject or Marks.
4. Click "Update Grade".

Changes are saved immediately.

Delete Grade

Purpose: Remove an unwanted subject record.

Steps:

1. Open View Students.
2. Click "Delete".
3. Confirm the deletion.

The record is permanently removed.

6. PROJECT DELIVERY & LINKS

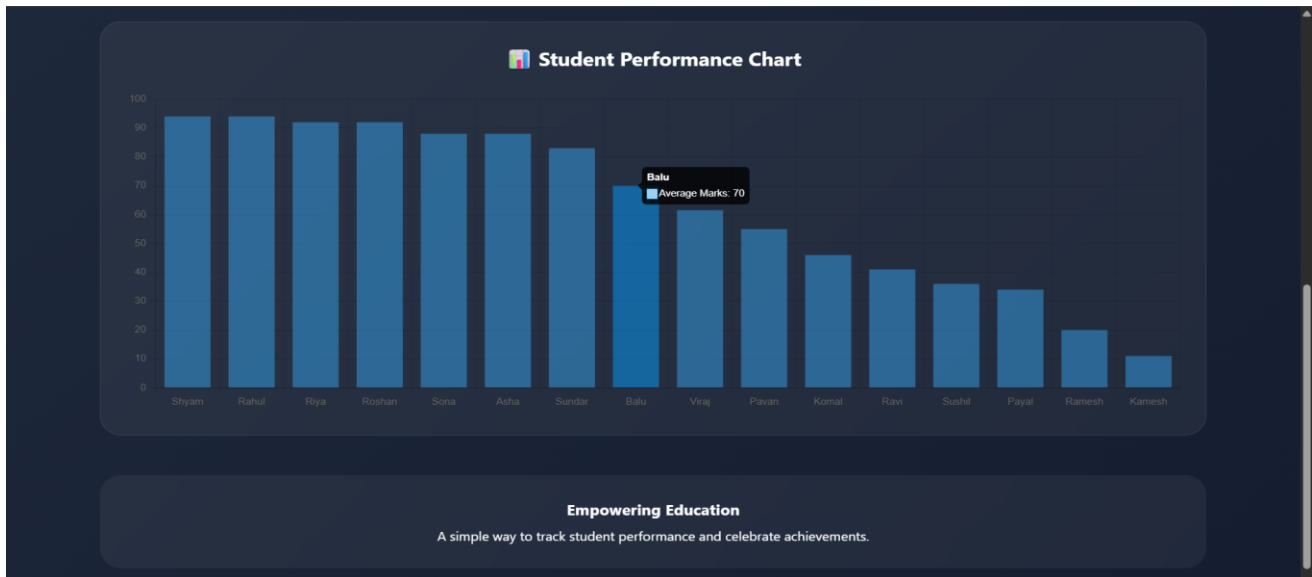
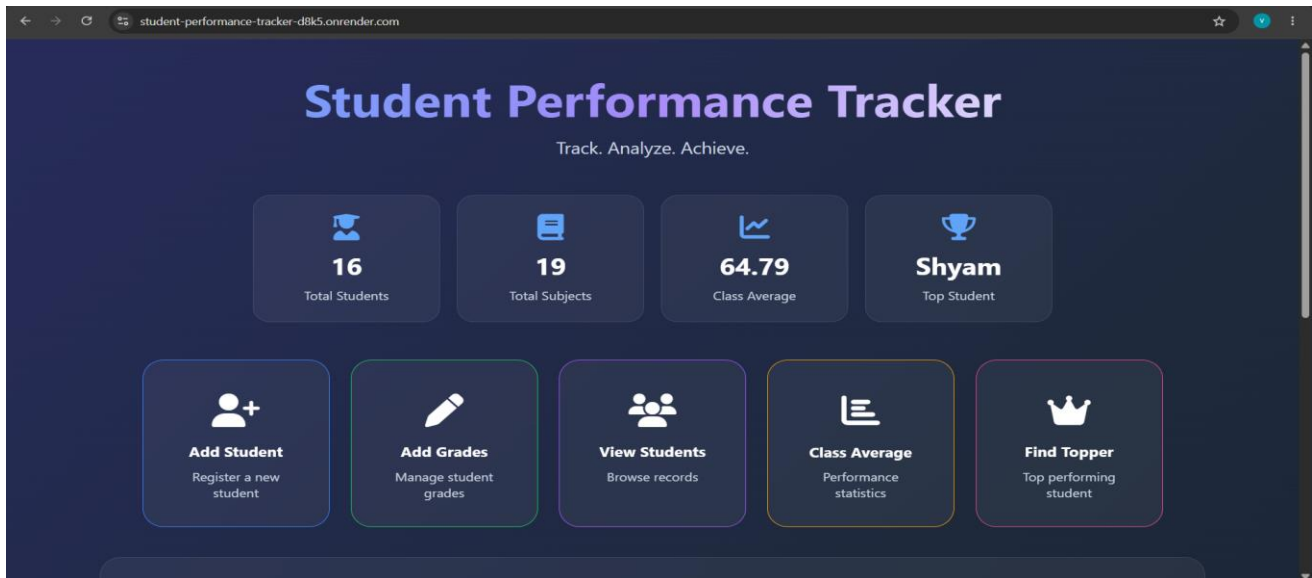
GitHub Repository Link

<https://github.com/baburao-analyst/student-performance-tracker>

Live Application Link

<https://student-performance-tracker-d8k5.onrender.com/>

Screenshots



Student Performance Tracker

Add Student

Add Student

[← Back to Home](#)

Student Performance Tracker

Add Grades

Add Grade

[← Back to Home](#)

Student Performance Tracker

View Students

Search by Roll Number or Name

Search

Roll Number	Name	Subject	Marks	Action
101	Pavan	Social	55.0	Report
				Edit
				Delete
102	Balu	GK	81.0	Report
				Edit

Student Report

Balu

Roll Number: 102

Subject	Marks
GK	81.0
Science	59.0

Average Marks: 70.0

[← Back to Students](#)

Student Performance Tracker

Edit Grade

Roll Number

102

Subject

GK

Marks

81.0

Update Grade

[← Back to Students](#)

Student Performance Tracker

⚠ Confirm Subject Deletion

Are you sure you want to delete this subject?

Roll Number: 102

Subject: GK

Marks: 81.0

Yes, Delete

[Cancel](#)

Student Performance Tracker

✓ **Success**

Student deleted successfully!

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Student Performance Tracker

Student Averages

Roll Number	Name	Average Marks
101	Pavan	55.00
102	Balu	70.00
103	Rahul	94.00
104	Viraj	61.50
105	Shyam	94.00
106	Ramesh	20.00
107	Kamesh	11.00
108	Komal	46.00
109	Ravi	41.00

Student Performance Tracker

Top Performer(s)

Name: Rahul

Roll Number: 103

Name: Shyam

Roll Number: 105

Average Marks: 94.0

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Conclusion

The Student Performance Tracker provides a simple and efficient way to manage academic records. The application supports record creation, viewing, updating, deletion, reporting, and performance analysis through an easy-to-use web interface.